1) Find 10 different model providers and their models released in last 12 months

Details to collect:

Model Provider | Model | Release Date | Cut off Date | Cost of tokens | Context Window | Type | When to use this model |

Model Provider	Model	Release Date	Cut off Date	Cost of tokens (Input/Output per 1M)	Context Window	Type	When to use this model
Anthropic	Claude Opus 5	July 24, 2026	May-26	$5.00 / $25.00	1,000,000	Frontier Multimodal	Best overall for complex reasoning, terminal use, and advanced agentic coding.
OpenAI	GPT-5.6 Sol	July 9, 2026	Feb 16, 2026	$5.00 / $30.00	1,050,000	Frontier Multimodal	Complex professional work, heavy reasoning, and logic-heavy workflows.
Google	Gemini 3.5 Flash	Mid 2026	Jan-25	$1.50 / $9.00	1,000,000	Multimodal	High-speed processing of large 1M context workflows and multimodal data.
DeepSeek	DeepSeek V4 Pro	Apr-26	Jan-26	$0.435 / $0.87	1,000,000	MoE Text & Code	Highly cost-effective infrastructure for large-context RAG pipelines.
Amazon	Amazon Nova 2 Lite	Dec-25	N/A	N/A	1,000,000	Text / Multimodal	Enterprise deployments integrated natively within the AWS ecosystem.
Moonshot AI	Kimi K3	Mid 2026	N/A	$3.00 / $15.00	1,048,576	Text LLM	Web browsing automations and deep,

							long-context document analysis.
Anthropic	Claude Mythos 5	Mid 2026	Jan-26	$10.00 / $50.00	1,000,000	Frontier Multimodal	Premium tier tasks that require extreme intelligence or logic-proving capabilities.
Alibaba	Qwen 3.7 Max	Mid 2026	2026	N/A	~984,000	Sparse MoE	Processing extensive multilingual datasets and building sovereign AI agents.
OpenAI	GPT-5.6 Luna	Mid 2026	Feb 16, 2026	$0.20 / $1.20	1,050,000	Cost-Optimized LLM	Budget-conscious, high-volume workloads needing full 1M context limits.
Zhipu AI	GLM 5.2	Mid 2026	Mar-26	$0.95 / $3.00	1,000,000	Text / Code	Very fast inference needs (347 tokens/sec) and broad general automations.

2) Read 2–3 articles about how transformer works?

Goal to understand:

(a) Self-attention mechanism

(b) Vector embedding

(c) Feed forward Nueral Network

1. "Illustrated Guide to Transformers" (by Michael Phi / Towards Data Science)

- **Why it's great:** This is widely considered one of the best visual introductions to the Transformer model. It takes you step-by-step through the encoder-decoder architecture.
- **Key Coverage:**
    - **Vector Embeddings:** Explains how input words are converted into continuous vector representations and how positional encodings are added to preserve word order.
    - **Self-Attention Mechanism:** Breaks down the query, key, and value vectors (Q, K, V) with clear visual diagrams to show how words relate to one another in context.
    - **Feed-Forward Networks:** Illustrates how the outputs from the multi-head attention layers are passed through position-wise fully connected feed-forward networks for further processing.

2. "The Illustrated Transformer" (by Jay Alammar)

- **Why it's great:** The gold standard for deep learning visualization. Jay Alammar's blog post uses intuitive color-coded diagrams that make complex matrix operations remarkably easy to grasp.
- **Key Coverage:**
    - **Vector Embeddings & Positional Encoding:** Clearly demonstrates the dimensionality of vectors and why positional information is crucial for sequence-based models.
    - **Self-Attention Mechanism:** Offers a brilliant, intuitive walkthrough of the self-attention calculation step-by-scale (including the scaling factor and softmax function).
    - **Feed-Forward Layers:** Explains the exact path data takes through the dense feed-forward sub-layers within each encoder and decoder block.

3. "Transformers, explained: The architectural breakthrough behind GPT and modern LLMs" (by Codecademy / Tech Blog or similar deep dive, alternatively: Hugging Face's "The Transformer Architecture" Chapter)

- **Why it's great:** A fantastic text-and-code synthesis that bridges theory with practical machine learning concepts.
- **Key Coverage:**

- **Vector Embeddings:** Details tokenization and embedding spaces where semantic meaning is captured geometrically.
- **Self-Attention & Multi-Head Attention:** Explains *why* multiple attention "heads" are necessary to capture different types of relationships simultaneously (e.g., grammatical vs. semantic).
- **Feed-Forward Neural Networks (FFN):** Covers the non-linear transformations (such as ReLU or GELU activations) applied after the attention pooling step to extract higher-level feature representations.

3) Connect with your group learners and connect with their LinkedIn profiles

This is done